

Hoeffding equation and PLA

The inputs for the tasks are in Numpy's npz format. They contain 4 arrays: `X`, `Y` for the training inputs and outputs, and `X_test` and `Y_test` for the test inputs and outputs, respectively. The samples are the rows of the `X` matrices.

Use only the training data for calculating the parameters (weights). Some input files contain reference solutions, you can use these to check the correctness of your answer (but only for that).

1. *Hoeffding equation*. You are testing a hypothesis on 43 samples, sampled from the data distribution. After measurement, you find your error to be 0.1. Your boss told you that she will tolerate at most an error of 0.2 in production (averaged for the long run). Can you bound the probability that the error will be larger (and she will be unhappy with you)?¹
2. Implement the Perceptron Learning Algorithm.
 - a. Find a linear separator that separates the two classes (`-1` and `1`) in the `pla.npz` dataset.
 - b. Plot the training data and the separator in each iteration. (You can use the code in `visualise.py`.)²

A simple way to do nonblocking plot in Matplotlib is:

```
import matplotlib.pyplot as plt
from visualise import visualise
fig, ax = None, None
while not done:
    w = calculate()
    fig, ax = visualise(w, X, Y, block=False, fig=fig, ax=ax)
    plt.pause(0.5)
```

- c. Repeat a) and b) on `pocket.npz` as well. How did it perform? Can you do better with a small modification?

¹The answer is "Yes, you can bound it"; the correct solution to this exercise includes your bound and its derivation.

²To plot an animation, you can use the guides here: https://matplotlib.org/stable/api/animation_api.html